Buttons in pygame

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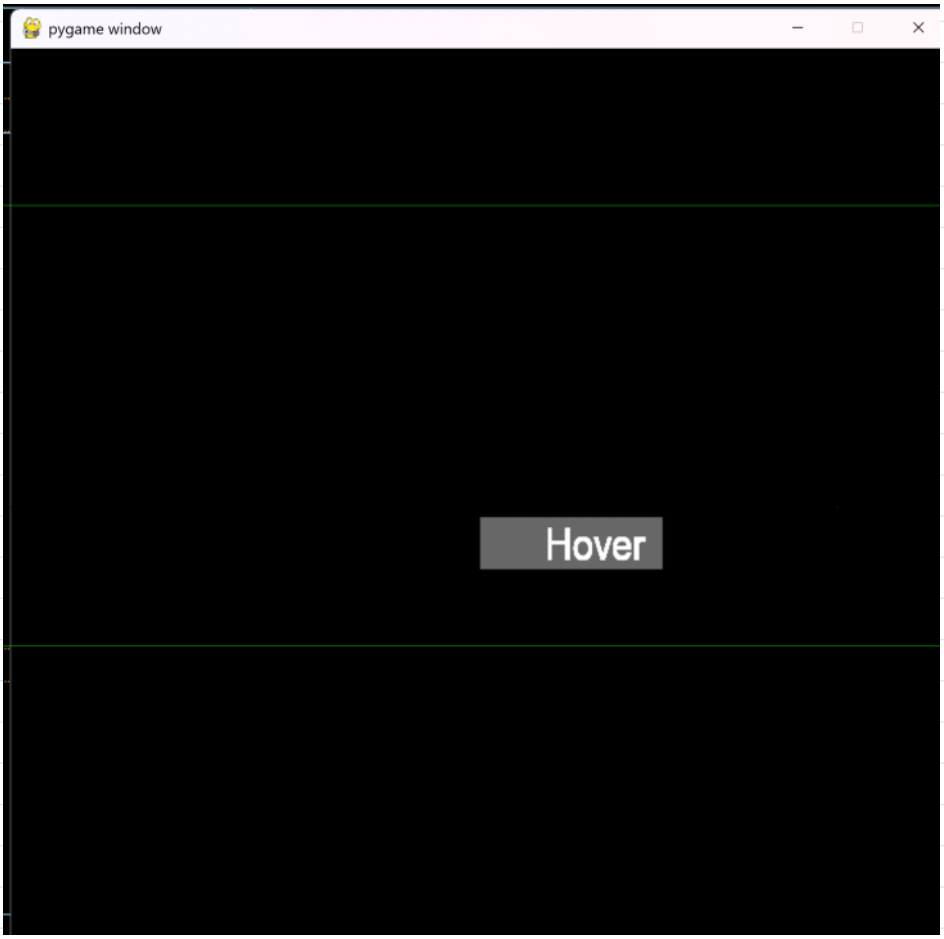
A game must have interactable buttons that can control different events in the game, to make the game more controlled and to add a proper GUI in it. These can be created in pygame by creating a rectangle onto the screen and then superimposing the indicating text on it. For this we will be using various functions like draw.rect(), screen.blit() etc. To add more liveliness to it, we can change the color of the button as the mouse has hovered on it. This can be done by using a function that updates the X and Y position of the mouse pointer and storing it in a tuple in a variable. Then we can set the boundaries of the rectangle into respective variables and check if the mouse is in those boundaries. If so, the color of the block will be changed to lighter shade to indicate that buttons is interactable.

Below is the implementation:

```
import pygame
import sys
pygame.init()
res = (720,720)
screen = pygame.display.set_mode(res)
color = (255,255,255) #white
color_light = (170,170,170) #light shade of the button
color_dark = (100,100,100) #dark shade of the button
width = screen.get_width() # Stores the width of the screen into a variable.
height = screen.get_height() # Stores the height of the screen into a variable.
smallfont = pygame.font.SysFont('Arial', 35) # Defining a font.
text = smallfont.render('Hover', True, color)
while True:
    for event in pygame.event.get():
        if event.type  == pygame.QUIT:
            pygame.quit()

        # Check if a mouse is clicked.
        mouse = pygame.mouse.get_pos()
        # Super imposing the text onto our button
        screen.blit(text,(width/2+50, height/2))
        if event.type == pygame.MOUSEBUTTONDOWN:
            # If the mouse is clicked the hovering effect we get
            if width/2 <= mouse[0] <= width/2 + 140 and height/2 <= mouse[1] <=
height/2 + 40:
                pygame.draw.rect(screen, color_light,[width/2, height/2,140, 40])
            else:
                pygame.draw.rect(screen, color_dark,[width/2, height/2,140,40])

        #update the frames of the game
        pygame.display.update()
```

Output:



Drawing design using arrow keys in PyGame

In PyGame with help of keys such that design that is marker moves horizontally when we are pressing the right arrow key or left arrow key on the keyboard and it moves vertically when we are pressing up arrow key or down arrow key. We can do this by making a spot(marker) on the respective coordinates which gets changes with the help of keys.

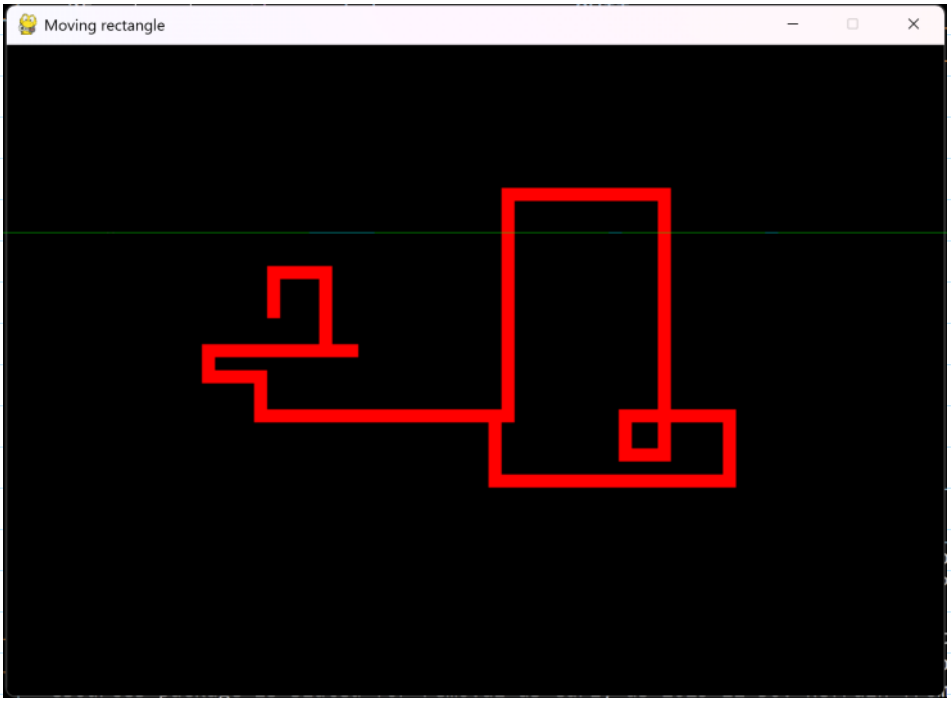
Change in coordinates of marker for respective key pressed:
Left arrow key: decrement in X coordinate.
Right arrow key: increment in X coordinate.
Up arrow key: decrement in Y coordinate.
Down arrow key: Increment in Y coordinate.

Below is the implementation:

```
import pygame
pygame.init()
win = pygame.display.set_mode((720, 500))
pygame.display.set_caption("Moving rectangle")
# Marker current coordinates.
x = 200
y = 200
```

```
# Dimension of the marker.
width = 10
height = 10
# Velocity or speed of movement?
vel = 10
while True:
    pygame.time.delay(10)

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit() # quiting the game

        keys = pygame.key.get_pressed()
        # If left arrow key is pressed.
        if keys[pygame.K_LEFT] and x > 0:
            x -= vel # Decrement in X coordinate.
        # If right arrow key is pressed.
        if keys[pygame.K_RIGHT] and x < 720 - width:
            x += vel #increment in x-coordinate
        # If up arrow key is pressed.
        if keys[pygame.K_UP] and y > 0:
            y -= vel # Decrement in Y coordinate.
        # If down arrow key is pressed.
        if keys[pygame.K_DOWN] and y < 500 - height:
            y += vel # Increment in Y coordinate.

        # Drawing spot on screen which is rectangle here.
        pygame.draw.rect(win, (255,0,0),(x,y,width, height))

        pygame.display.update() # It refreshes the window.
```

Output:



Home work

Question 1: Interactive Start Button
Write a PyGame program that creates a **"START" button** in the center of the screen.
Requirements:
1. Display a rectangular button labeled **START.**
2. When the mouse **hovers** over the button, its color should change.
3. When the user **clicks** the button:
    - Print "Game Started" in the console.
    - Change the window caption to "Game Running".
4. Use pygame.draw.rect() and screen.blit().

Question 2: Keyboard Controlled Player Box
Create a PyGame program where a **player box** moves using arrow keys.
Requirements:
1. Draw a square of size **20x20.**
2. Control movement using:
    - LEFT → move left
    - RIGHT → move right
    - UP → move up
    - DOWN → move down
3. The box must **not move outside** the screen.
4. Display the **current X and Y coordinates** on the screen.

Question 3: Button Controlled Movement
Design a PyGame program with **four buttons**:
- UP
- DOWN
- LEFT
- RIGHT

Requirements:

1. Place buttons at the bottom of the screen.
2. Clicking each button should move a rectangle accordingly.
3. Hovering over any button should change its color.
4. Movement speed should be **10 pixels per click**.
5. The rectangle should stay inside the window boundaries.